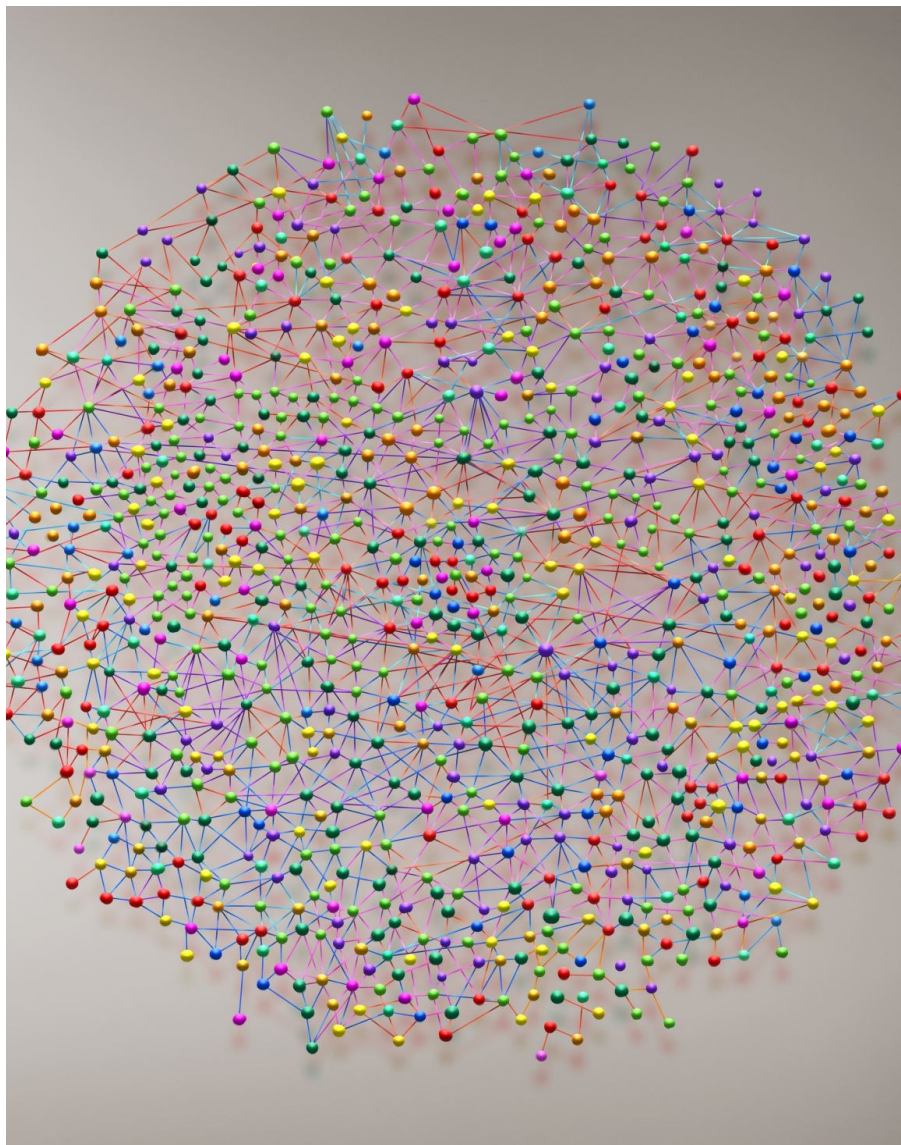


Lecture 25 – D-Separation

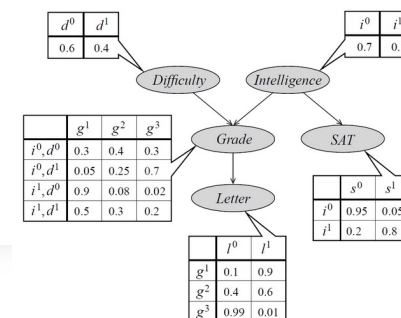
Prof Xiaowei Huang

<https://cgi.csc.liv.ac.uk/~xiaowei/>

(Attendance Code: **864387**)

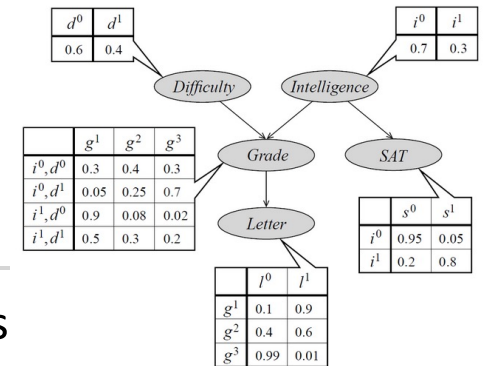


Up to now,



- Traditional Machine Learning Algorithms
- Adversarial Attack and Defence
- Deep learning
- Probabilistic Graphical Models
 - Introduction
 - Distribution $\Leftrightarrow$ Graph
 - Reasoning Patterns (Causal Reasoning, Evidential Reasoning, Intercausal Reasoning)

Recap: Local Independencies



- Graph G with CPDs is equivalent to a set of independence assertions

$$P(D, I, G, S, L) = P(D)P(I)P(G|D, I)P(S|I)P(L|G)$$

- Local Conditional Independence Assertions (starting from leaf nodes):

$$I_{local}(G) = \{ (L \perp I, D, S | G), \quad L \text{ is conditionally independent of all other nodes given parent } G \\ (S \perp D, G, L | I), \quad S \text{ is conditionally independent of all other nodes given parent } I \\ (G \perp S | D, I), \quad \text{Even given parents, } G \text{ is NOT independent of descendant } L \\ (I \perp D | \emptyset), \quad \text{Nodes with no parents are marginally independent} \\ (D \perp I, S | \emptyset) \} \quad D \text{ is independent of non-descendants } I \text{ and } S$$

- Parents of a variable shield it from probabilistic influence
 - Once value of parents known, no influence of ancestors
- Information about descendants can change beliefs about a node

- $I_{local}(G) \subseteq I(G)$

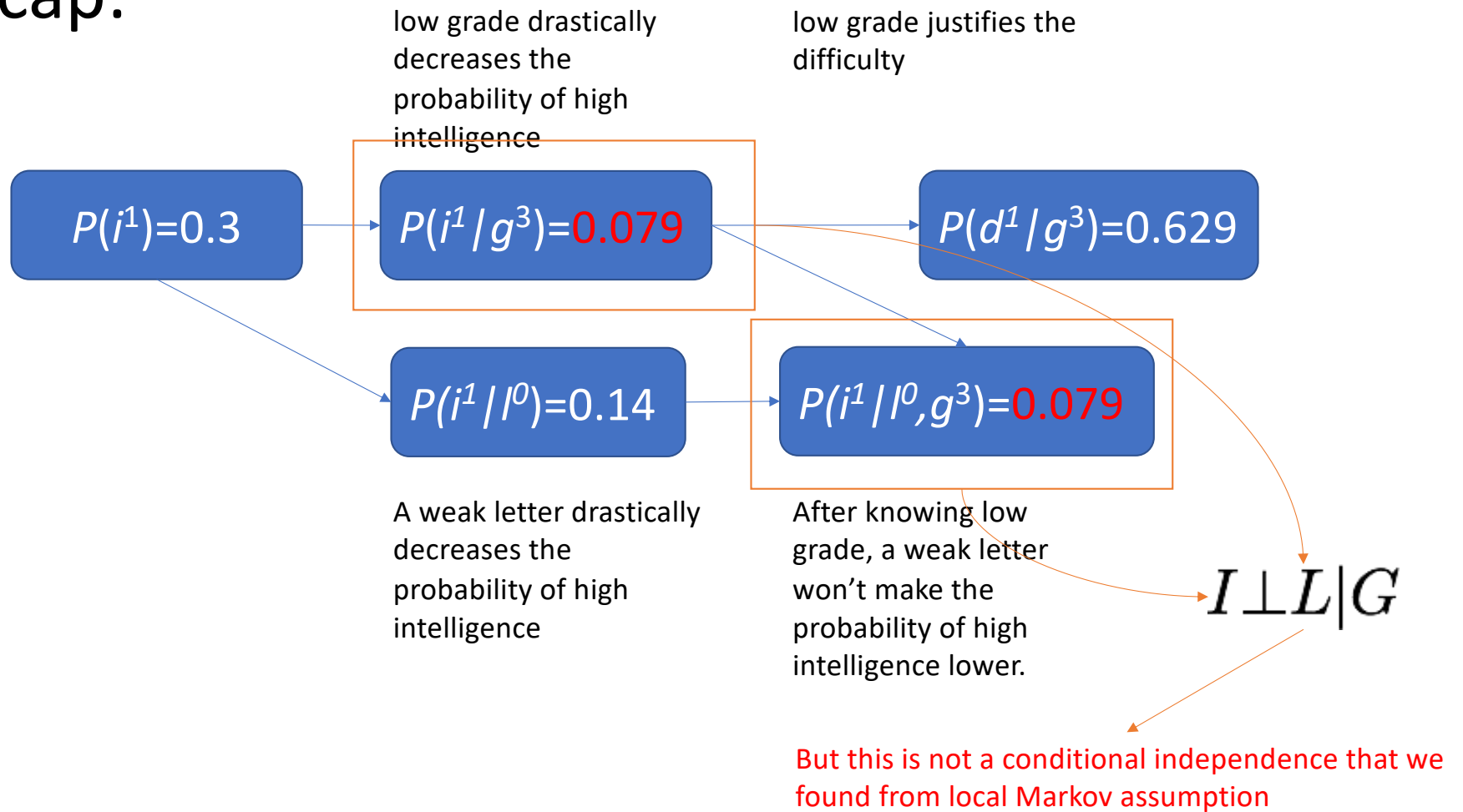
Can we have the following conditional independence?

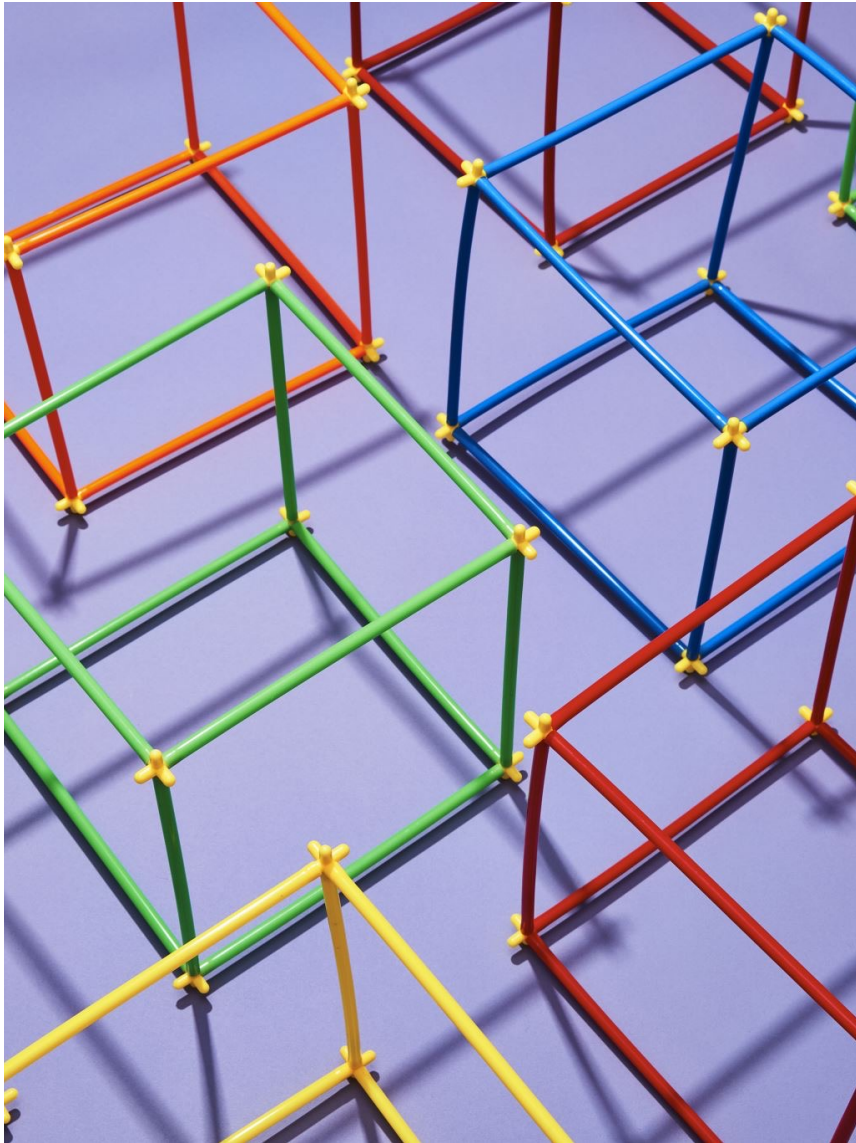
$$D \perp S | G$$

$$D \perp S | I$$

$$D \perp S | G, I$$

Recap:



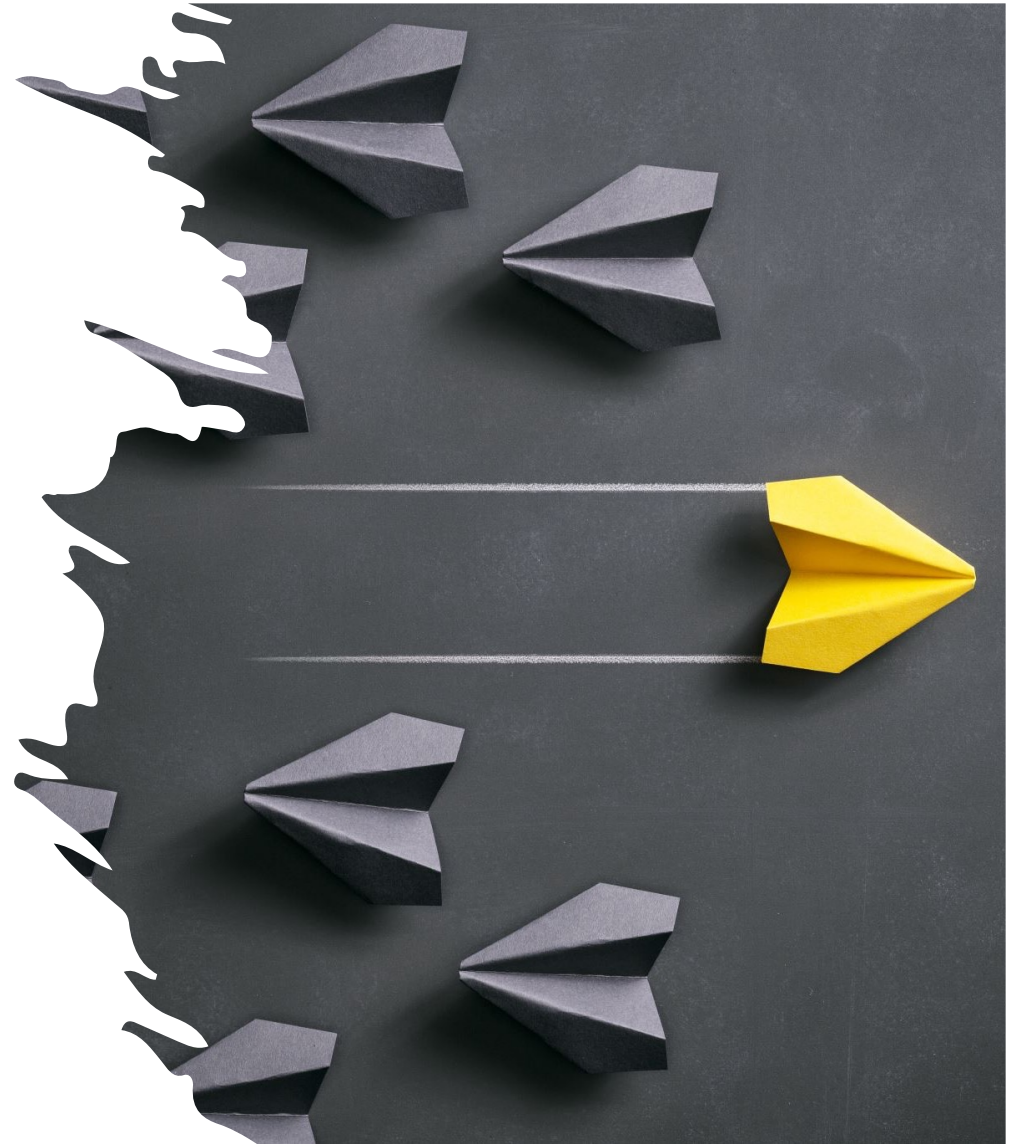


Independencies in Graphs

- A graph structure G encodes a set of conditional independence assumptions $I(G)$
- Are there other independencies that we can read-off?
 - i.e., are there other independencies that hold for every distribution that factorizes over G ?
- D-separation holds the key

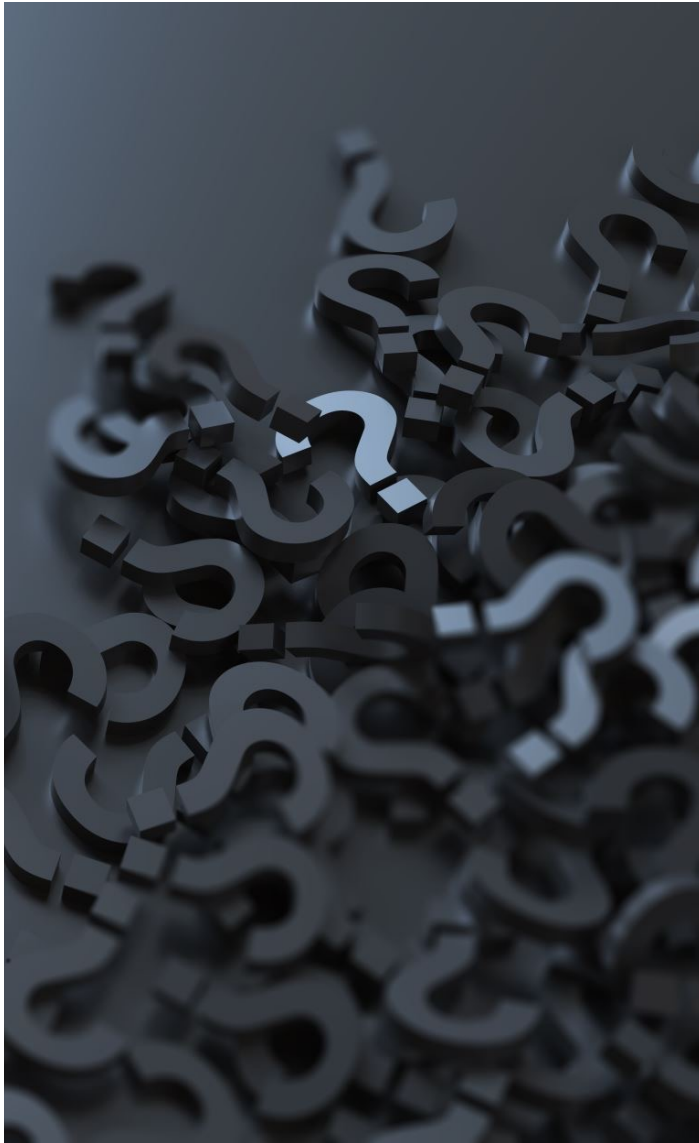
Topics

- Why D-separation?
- What is D-separation?
- Algorithm for D-separation (extended materials)
- More Example of D-separation





Why D-separation?



Dependencies and Independencies

- Crucial for understanding network behaviour
- Independence properties are important for answering queries
 - Exploited to reduce computation of inference
 - A distribution P that factorizes over G satisfies $I(G)$
- *Local Markov assumption cannot figure out all conditional independencies*



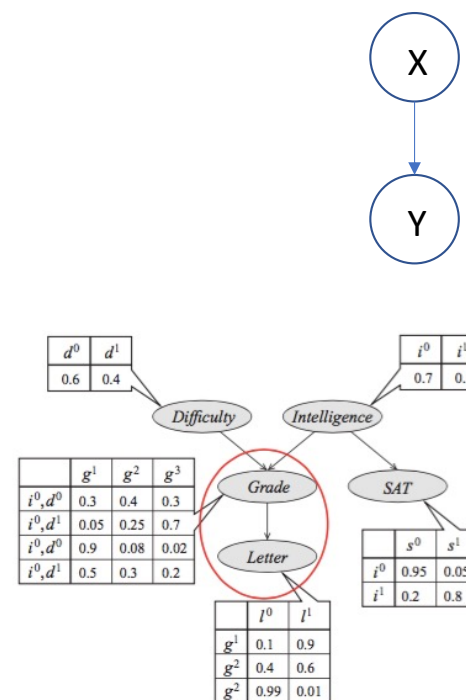
What is D-separation?

D-separation

- Study independence properties for subgraphs (**connected triples**)
- Analyze complex cases in terms of triples along paths between variables
- **D-separation**: a condition / algorithm for answering such queries
- **Definition**: A procedure $\text{d-sep}_G(X \perp Y | Z)$ that given a DAG G , and sets X , Y , and Z returns either *yes* or *no*, where $\text{d-sep}_G(X \perp Y | Z) = \text{Yes}$ iff $(X \perp Y | Z)$ follows from $I(G)$

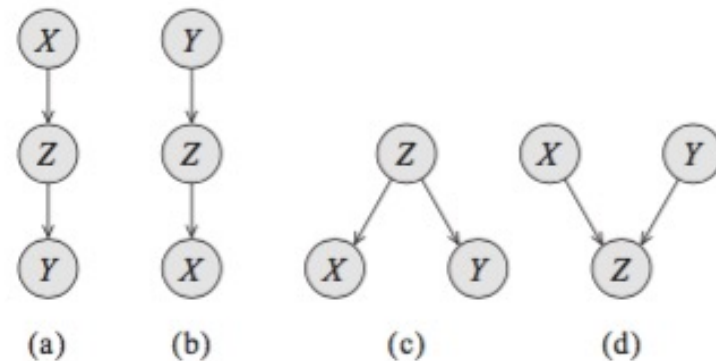
Direct Connection between X and Y

- X and Y are correlated regardless of any evidence about any other variables
 - E.g., Feature Y and character X are correlated
 - Grade G and Letter L are correlated
- If X and Y are directly connected we can get examples where they influence each other regardless of Z



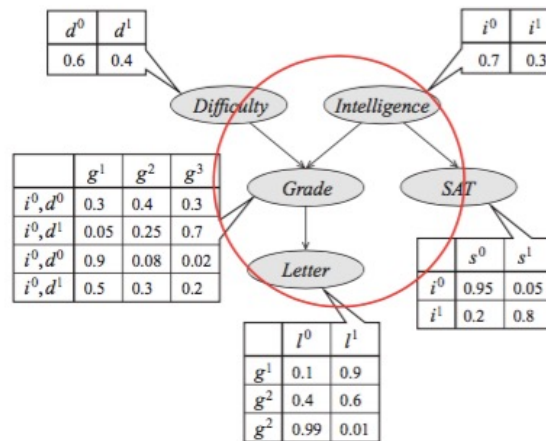
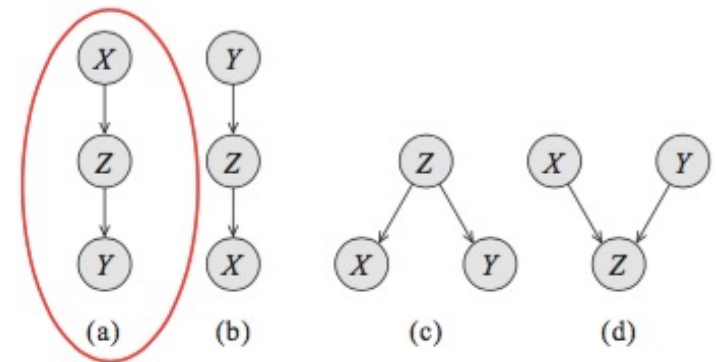
Indirect Connection between X and Y

- Four cases where X and Y are connected via Z
 - (a). Indirect causal effect
 - (b). Indirect evidential effect
 - (c). Common cause
 - (d). Common effect
- We will see that first three cases are similar while fourth case (V-structure) is different



1. Indirect Causal Effect: $X \rightarrow Z \rightarrow Y$

- Cause X cannot influence effect Y if Z observed
 - Observed Z blocks influence
- If *Grade* observed then I does not influence L
 - *Intelligence* influences *Letter* if *Grade* is unobserved



$Z = \text{Grade}$
 $I \perp\!\!\!\perp L \mid G$

Recap:

low grade drastically decreases the probability of high intelligence

low grade justifies the difficulty

$$P(i^1)=0.3$$

$$P(i^1 | g^3)=0.079$$

$$P(d^1 | g^3)=0.629$$

$$P(i^1 | l^0)=0.14$$

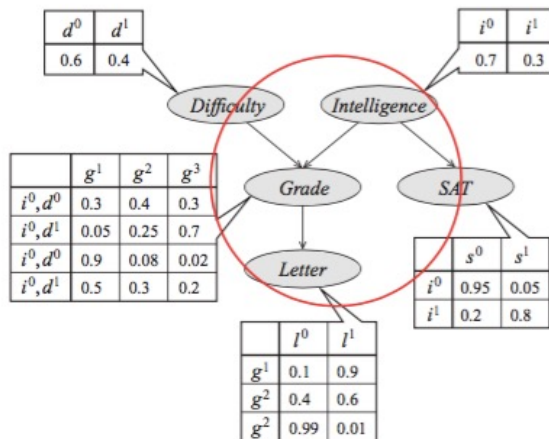
$$P(i^1 | l^0, g^3)=0.079$$

A weak letter drastically decreases the probability of high intelligence

After knowing low grade, a weak letter won't make the probability of high intelligence lower.

$$I \perp L | G$$

The indirect causal effect explains this conditional independence



Causal Chains

Evidence along the chain “blocks” the influence (makes “inactive”)

- This configuration is a “causal chain”

Guaranteed X independent of Z given Y?



X: Low pressure

Y: Rain

Z: Traffic

$$P(x, y, z) = P(x)P(y|x)P(z|y)$$

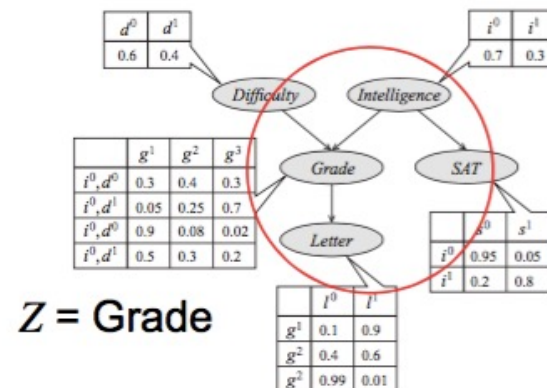
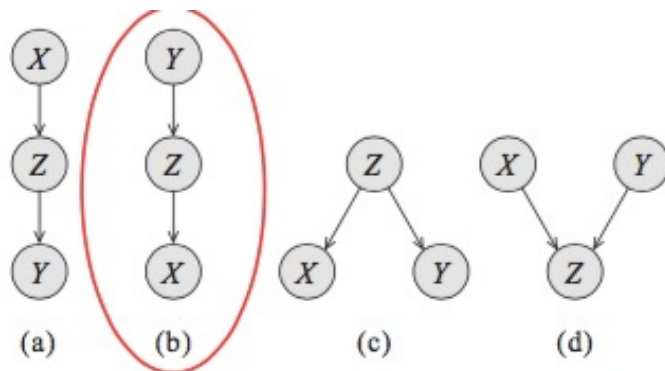
Indirect causal effect

$$\begin{aligned} P(z|x, y) &= \frac{P(x, y, z)}{P(x, y)} \\ &= \frac{\cancel{P(x)}\cancel{P(y|x)}P(z|y)}{\cancel{P(x)}\cancel{P(y|x)}} \\ &= P(z|y) \end{aligned}$$

Yes! x and z are conditionally Independent upon y

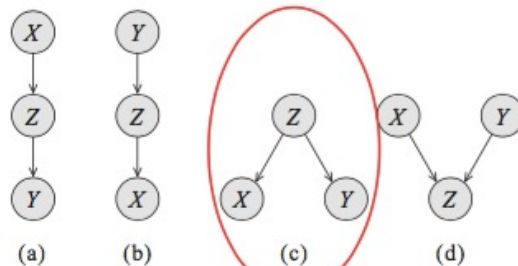
2. Indirect Evidential Effect: $Y \rightarrow Z \rightarrow X$

- Evidence X can influence Y via Z only if Z is unobserved
 - Observed Z blocks influence
- If Grade unobserved, Letter influences assessment of Intelligence
- Dependency is a symmetric notion
 - $X \perp Y$ does not hold then $Y \perp X$ does not hold either

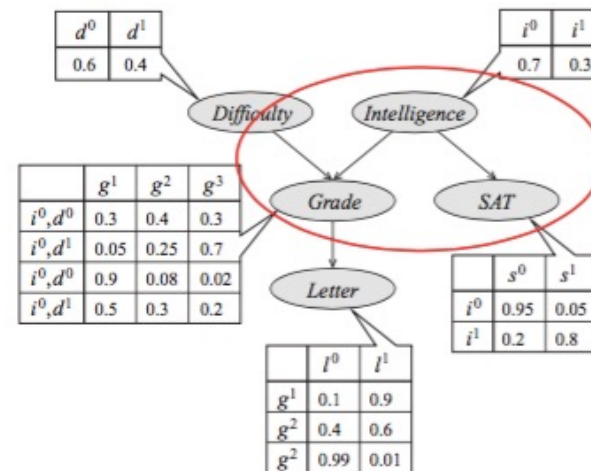


3. Common Cause: $X \leftarrow Z \rightarrow Y$

- X can influence Y if and only if Z is not observed
 - Observed Z blocks
- Grade is correlated with SAT score
- But if Intelligence is observed then SAT provides no additional information



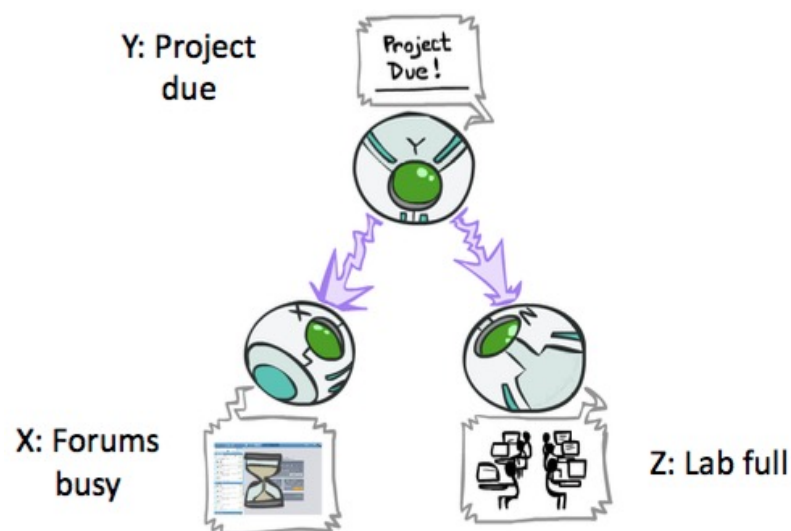
$$(S \perp G | I)$$



Common Cause

Observing the cause blocks influence between effects. (makes inactive)

- This configuration is a “common cause”



$$P(x, y, z) = P(y)P(x|y)P(z|y)$$

Common cause

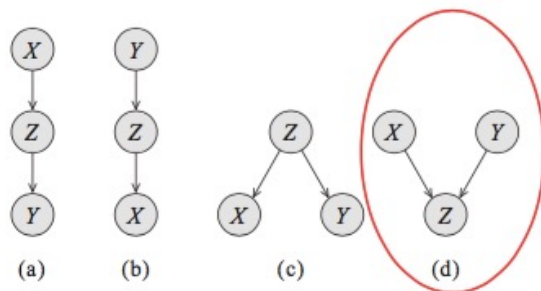
- Guaranteed X and Z independent given Y?

$$\begin{aligned} P(z|x, y) &= \frac{P(x, y, z)}{P(x, y)} \\ &= \frac{\cancel{P(y)}\cancel{P(x|y)}P(z|y)}{\cancel{P(y)}\cancel{P(x|y)}} \\ &= P(z|y) \end{aligned}$$

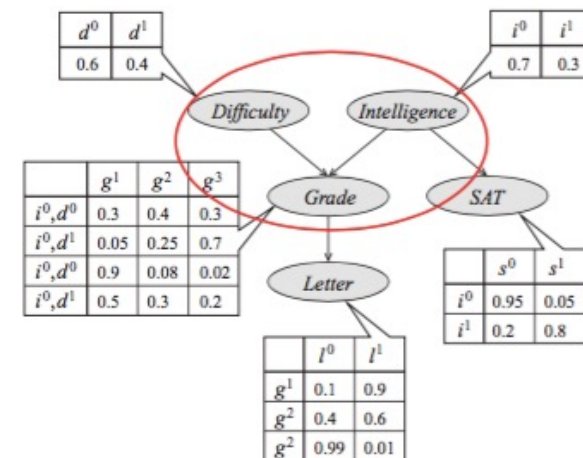
Yes! x and z are conditionally Independent upon y

4. Common Effect (V-structure) $X \rightarrow Z \leftarrow Y$

- Influence cannot flow on trail $X \rightarrow Z \leftarrow Y$ if Z is not observed
 - Observed Z enables
 - Opposite to previous 3 cases (Observed Z blocks)
- When G not observed I and D are independent
- When G is observed, I and D are correlated

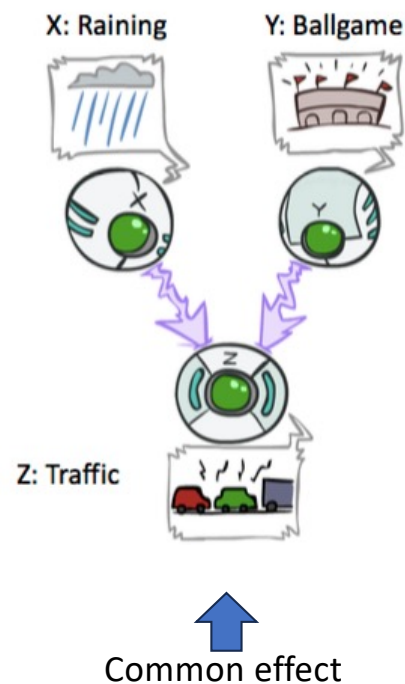


$$I \perp D | \sim G$$



Common Effect

- Last configuration: two causes of one effect (v-structures)

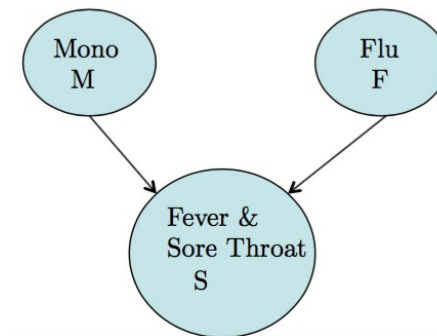


- Are X and Y independent?
 - **Yes**: the ballgame and the rain cause traffic, but they are not correlated
 - Still need to prove they must be (try it!)
- Are X and Y independent given Z?
 - **No**: seeing traffic puts the rain and the ballgame in competition as explanation.
- This is **backwards** from the other cases
 - Observing an effect **activates** influence between possible causes. (makes active!)

Recall: Common in Human Reasoning

- Binary Variables
- Fever & Sore Throat can be caused by mono and flu
- When flu is diagnosed probability of mono is reduced (although mono could still be present)
- It provides an alternative explanation of symptoms

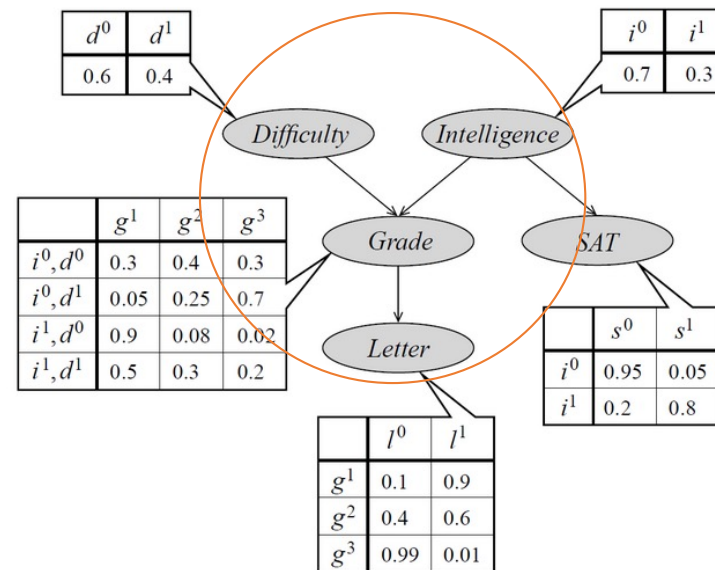
$$P(m^1/s^1) > P(m^1/s^1, f^1)$$



4. Common Effect (V-structure) $X \rightarrow Z \leftarrow Y$

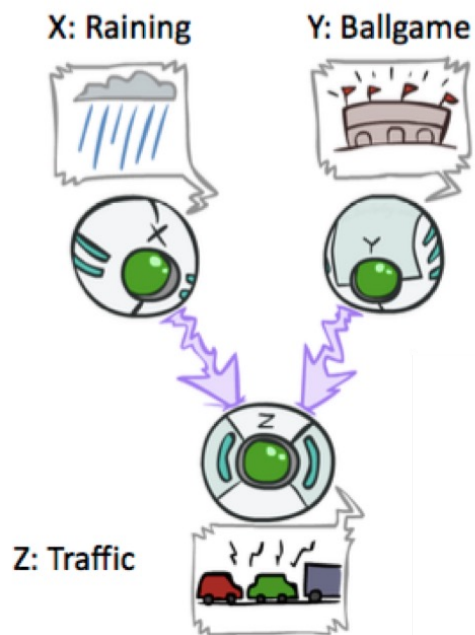
- *Grade* is not observed
- Observe weak letter *L*
 - Which indicates low *Grade*
 - Suffices to correlate *D* and *I*

Effective for not only child,
but also descendants



Example: Common Effect – why V-structure?

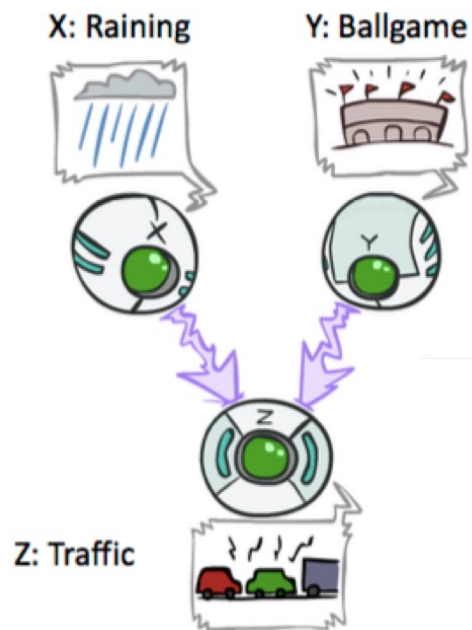
$$P(X) = 0.8$$



X	Y	Z	P
T	T	T	0.076
T	T	F	0.004
T	F	T	0.576
T	F	F	0.144
F	T	T	0.162
F	T	F	0.002
F	F	T	0.090
F	F	F	0.009

Example: Common Effect – why V-structure?

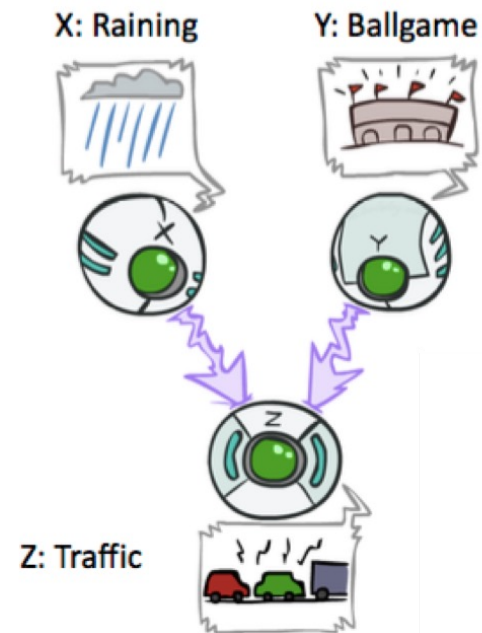
$$P(X) = 0.8$$



X	Y	Z	P
T	T	T	0.076
T	T	F	0.004
T	F	T	0.576
T	F	F	0.144
F	T	T	0.162
F	T	F	0.002
F	F	T	0.090
F	F	F	0.009

Example: Common Effect – why V-structure?

$$P(X) = 0.8$$



X	Y	Z	P
T	T	T	0.076
T	T	F	0.004
T	F	T	0.576
T	F	F	0.144
F	T	T	0.018
F	T	F	0.002
F	F	T	0.090
F	F	F	0.009

$$\begin{aligned} P(X|Y) &= \frac{0.076+0.004}{0.076+0.004+0.018+0.002} \\ &= 0.08 / 0.1 \\ &= 0.8 \end{aligned}$$

X and Y are
independent!

But Suppose Also Know $Z=T$ – why V-structure?

$$P(X) = 0.8$$

X: Raining



Y: Ballgame



Z: Traffic

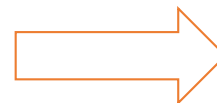


$$P(X|Z) = \frac{.076 + .576}{.076 + .576 + .018 + .090} = \frac{0.652}{0.76} = 0.858$$

$$P(X|Y,Z) = \frac{0.076}{0.076 + 0.018} = 0.8085$$

X	Y	Z	P
T	T	T	0.076
T	T	F	0.004
T	F	T	0.576
T	F	F	0.144
F	T	T	0.018
F	T	F	0.002
F	F	T	0.090
F	F	F	0.009

X and Y are not independent given Z!



So, observing Z makes X and Y dependent

Summary of Indirect Connection

- Causal trail: $X \rightarrow Z \rightarrow Y$: active iff Z not observed
- Evidential Trail: $X \leftarrow Z \leftarrow Y$: active iff Z is not observed
- Common Cause: $X \leftarrow Z \rightarrow Y$: active iff Z is not observed
- Common Effect: $X \rightarrow Z \leftarrow Y$: active iff either Z or one of its descendants is observed

What is the general case?

The General Case

- General question: in a given BN, are two variables independent (given evidence)?
- Solution: analyze the graph
- Any complex example can be broken into repetitions of the three canonical cases

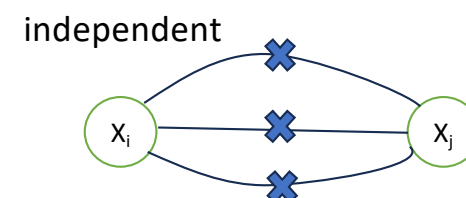
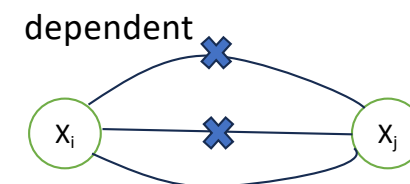
D-Separation

- Query: $X_i \perp X_j | \{X_{k_1}, \dots, X_{k_n}\}$?
- Check all (undirected) paths between X_i and X_j
 - If one or more paths is **active**, then independence not guaranteed

$$X_i \not\perp X_j | \{X_{k_1}, \dots, X_{k_n}\}$$

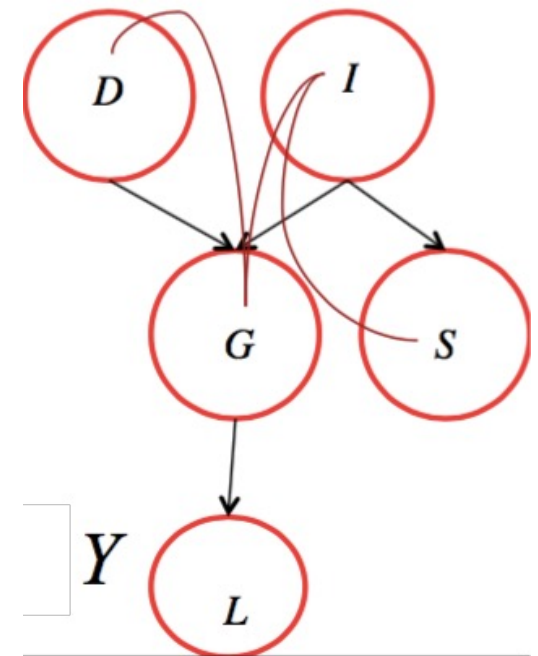
- Otherwise (i.e. if **all paths** are inactive),
then “D-separated” = independence **is** guaranteed

$$X_i \perp X_j | \{X_{k_1}, \dots, X_{k_n}\}$$



Active Trail

- When influence can flow from X to Y via Z then trail $X—Z—Y$ is active
- Example: Consider Trail $D \rightarrow G \leftarrow I \rightarrow S$
- When observed
 - $Z = \{\emptyset\}$: trail is *inactive* because v-structure $D \rightarrow G \leftarrow I$ is inactive
 - $Z = \{L\}$: *active* ($D \rightarrow G \leftarrow I$ active) since L is descendant of G
 - $Z = \{L, I\}$: *inactive* because observing I blocks $G \leftarrow I \rightarrow S$



D-separation definition

- Let $\mathbf{X}, \mathbf{Y}$ and $\mathbf{Z}$ be three sets of nodes in G .
- $\mathbf{X}$ and $\mathbf{Y}$ are d-separated given $\mathbf{Z}$ denoted $\text{d-sep}_G(\mathbf{X} \perp \mathbf{Y} | \mathbf{Z})$ if there is no active trail between any node $X \in \mathbf{X}$ and $Y \in \mathbf{Y}$ given $\mathbf{Z}$
- That is, nodes in $\mathbf{X}$ cannot influence nodes in $\mathbf{Y}$
- Provides notion of separation between nodes in a directed graph (“directed” separation)

Independencies from D-separation

- Definition

$$I(G) = \{(X \perp Y | Z) : \text{d-sep}_G(X \perp Y | Z)\}$$

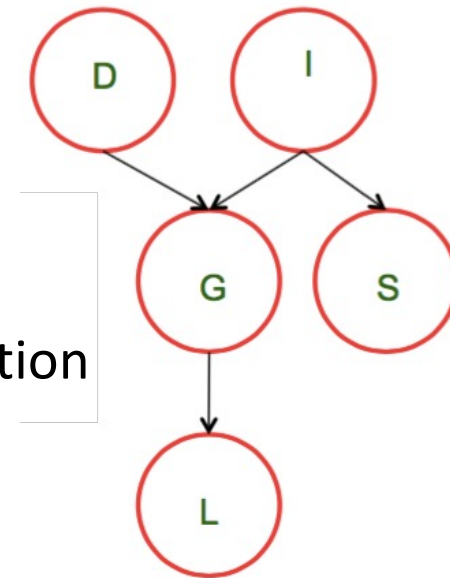
- Also called *Global Markov independencies*
- Note: Derived purely from graph (using trails)

- Example: *Global* independence using D-separation

- Compare with *local* independencies

- For each X_i : $(X_i \perp \text{Non-Desc}(X_i) \mid \text{Pa}(X_i))$

$$\{(L \perp I, D, S | G), (S \perp D, G, L | I), (G, S | D, I), (D \perp I, S | \emptyset)\} \subseteq I(G)$$

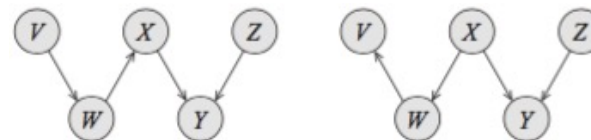




I-Equivalence

I-Equivalence

- Conditional independence assertion statements can be the same with different structures
- Two graphs G_1 and G_2 are *I-equivalent* if $I(G_1)=I(G_2)$
- Skeleton of a BN graph G is an undirected graph with an edge for every edge in G
- If two BN graphs have the same set of skeletons and v-structures then they are *I-equivalent*



Same skeleton

Same v-structure $X \rightarrow Y \leftarrow Z$

Bayes Nets Representation Summary

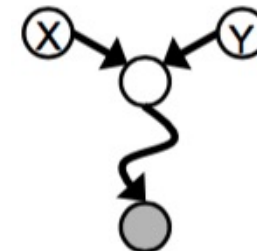
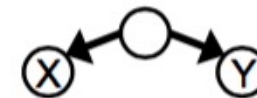
- Bayes nets compactly encode joint distributions
- Guaranteed independencies of distributions can be deduced from BN graph structure
- D-separation gives precise conditional independence guarantees from graph alone
- A Bayes' net's joint distribution may have further (conditional) independence that is not detectable until you inspect its specific distribution

More Example of D-separation

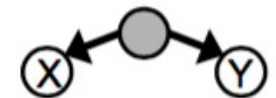
Active / Inactive Paths

- Question: Are X and Y conditionally independent given evidence variables {Z}?
 - Yes, if X and Y “d-separated” by Z
 - Consider all (undirected) paths from X to Y
 - If no path is active -> independence!
- A **path** is active if every triple in path is active:
 - Causal chain $A \rightarrow B \rightarrow C$ where B is unobserved (either direction)
 - Commoncause $A \leftarrow B \rightarrow C$ where B is unobserved
 - Common effect (aka v-structure)
 $A \rightarrow B \leftarrow C$ where B or one of its descendants is observed
- All it takes to block a path is a **single** inactive segment
 - (But **all** paths must be inactive)

Active Triples



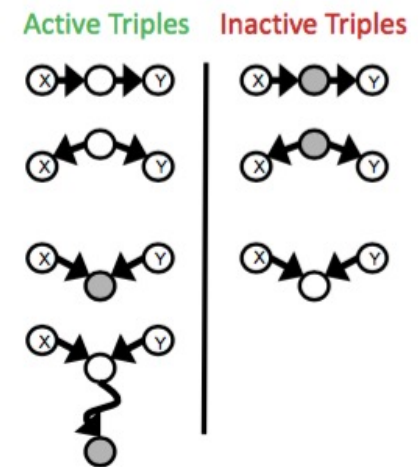
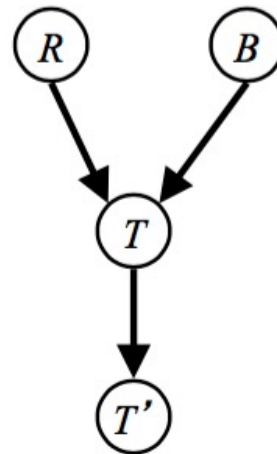
Inactive Triples



Example

$R \perp\!\!\!\perp B$

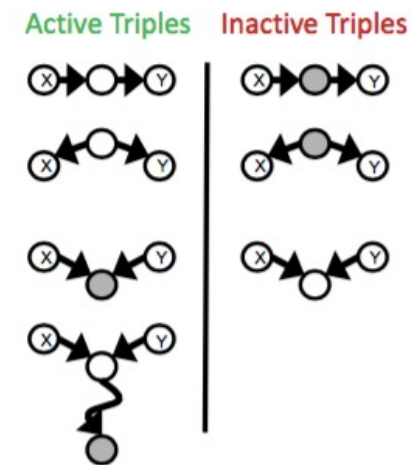
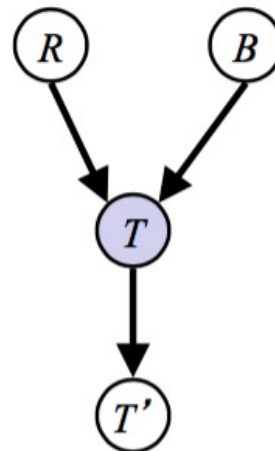
Yes, Independent!



Example

$R \perp\!\!\!\perp B$ *Yes, Independent!*

$R \perp\!\!\!\perp B|T$ *No*

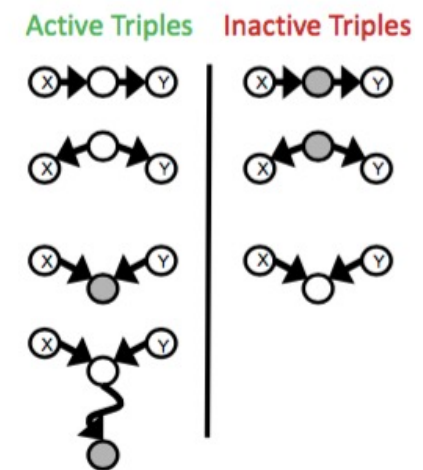
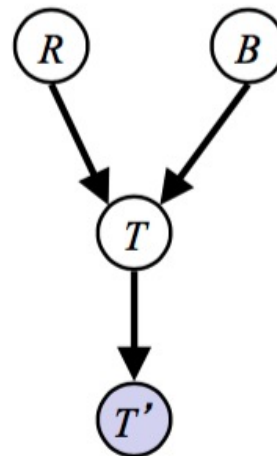


Example

$R \perp\!\!\!\perp B$ *Yes, Independent!*

$R \perp\!\!\!\perp B|T$ *No*

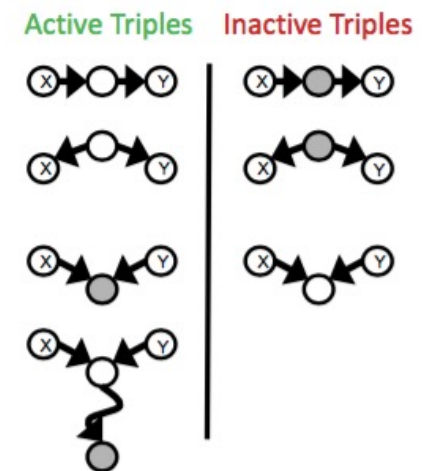
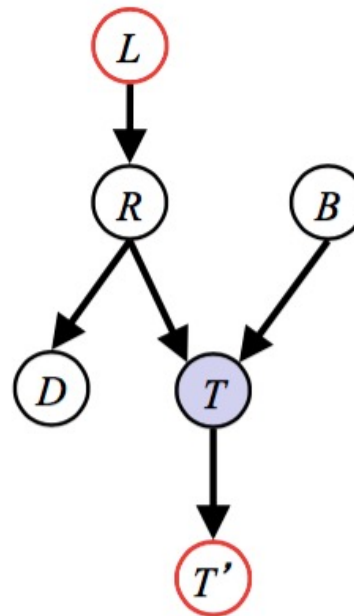
$R \perp\!\!\!\perp B|T'$ *No*



Example

$$L \perp\!\!\!\perp T' | T$$

Yes, Independent



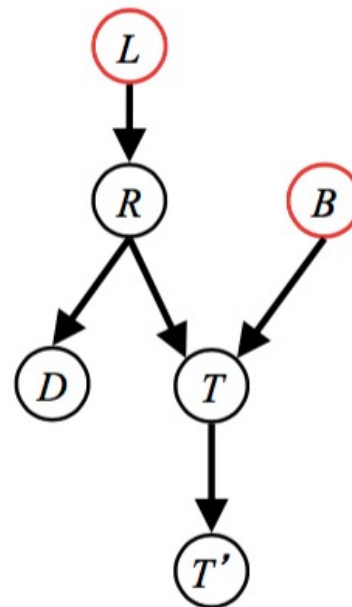
Example

$$L \perp\!\!\!\perp T' | T$$

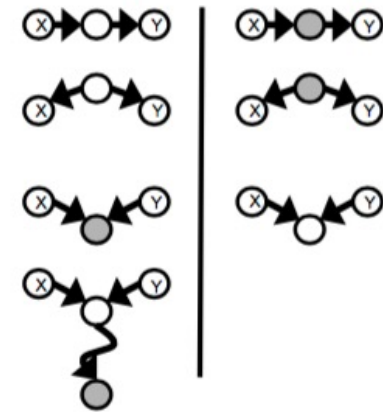
Yes, Independent

$$L \perp\!\!\!\perp B$$

Yes, Independent



Active Triples Inactive Triples



Example

$$L \perp\!\!\!\perp T' | T$$

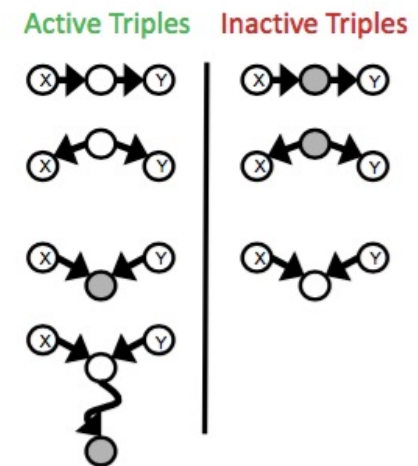
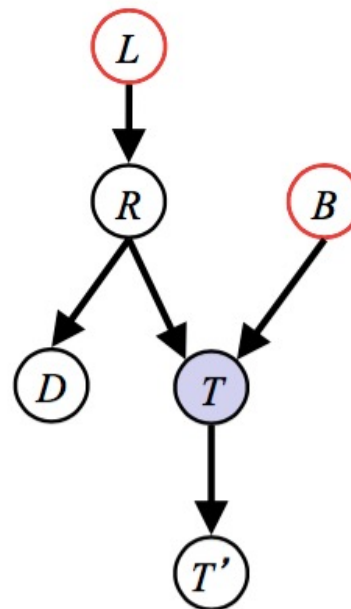
Yes, Independent

$$L \perp\!\!\!\perp B$$

Yes, Independent

$$L \perp\!\!\!\perp B | T$$

No



Example

$$L \perp\!\!\!\perp T' | T$$

Yes, Independent

$$L \perp\!\!\!\perp B$$

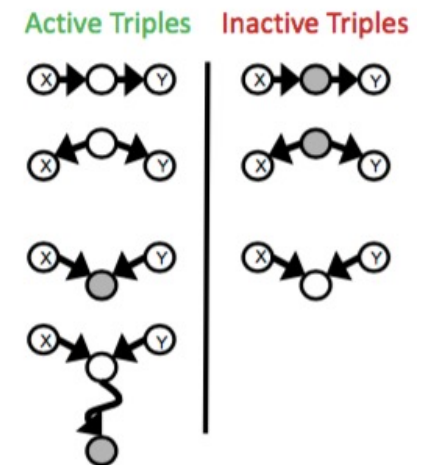
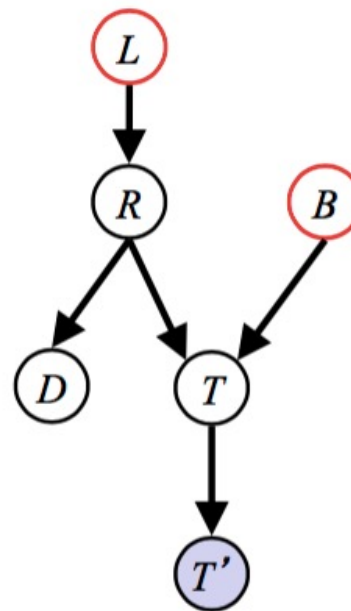
Yes, Independent

$$L \perp\!\!\!\perp B | T$$

No

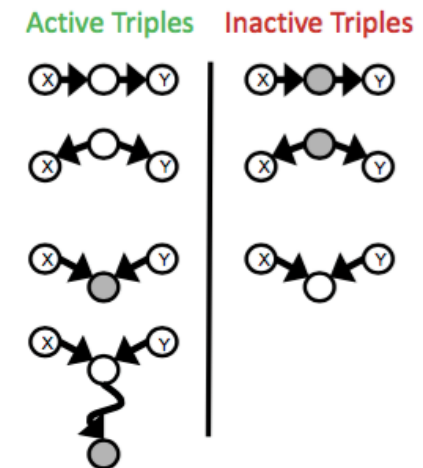
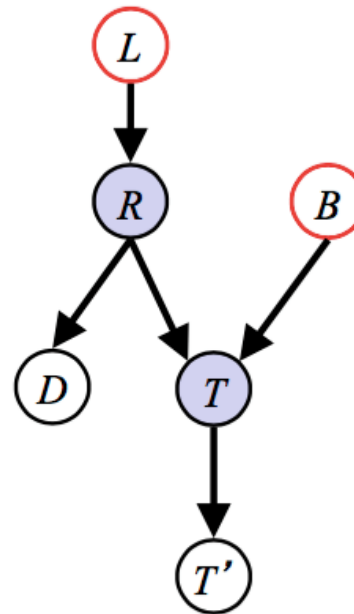
$$L \perp\!\!\!\perp B | T'$$

No



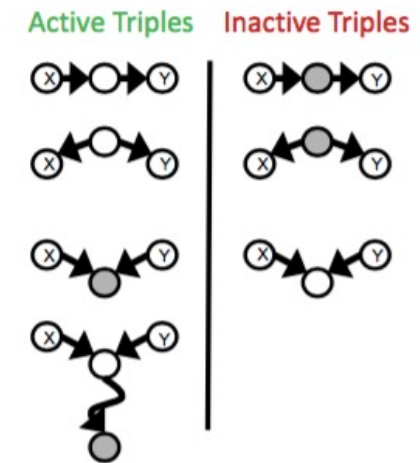
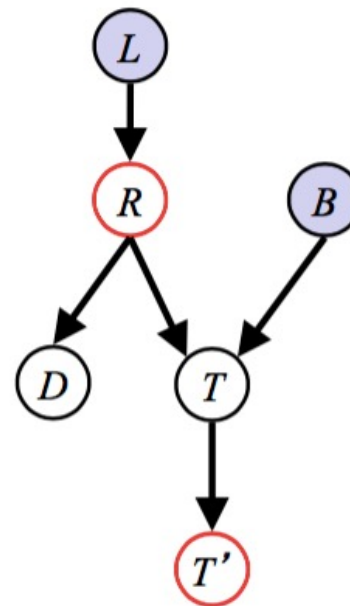
Example

$L \perp\!\!\!\perp T' T$	<i>Yes, Independent</i>
$L \perp\!\!\!\perp B$	<i>Yes, Independent</i>
$L \perp\!\!\!\perp B T$	<i>No</i>
$L \perp\!\!\!\perp B T'$	<i>No</i>
$L \perp\!\!\!\perp B T, R$	<i>Yes, Independent</i>



Example

$L \perp\!\!\!\perp T' T$	<i>Yes, Independent</i>
$L \perp\!\!\!\perp B$	<i>Yes, Independent</i>
$L \perp\!\!\!\perp B T$	<i>No</i>
$L \perp\!\!\!\perp B T'$	<i>No</i>
$L \perp\!\!\!\perp B T, R$	<i>Yes, Independent</i>
$R \perp\!\!\!\perp T' L, B$	<i>No</i>



Example

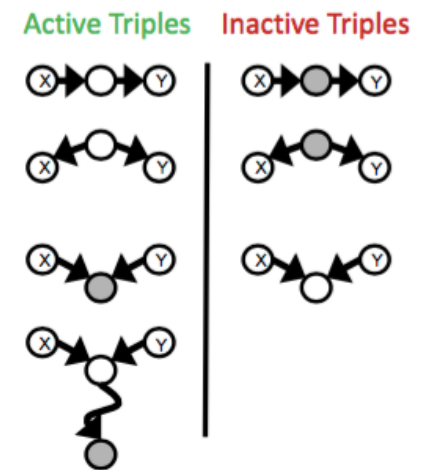
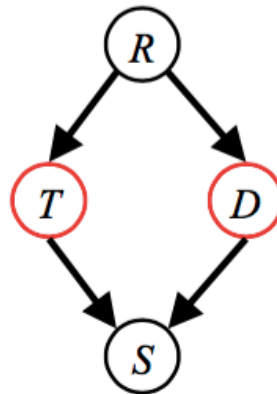
- Variables:

- R: Raining
- T: Traffic
- D: Roof drips
- S: I'm sad

- Questions:

$$T \perp\!\!\!\perp D$$

No



Example

- Variables:

- R: Raining
- T: Traffic
- D: Roof drips
- S: I'm sad

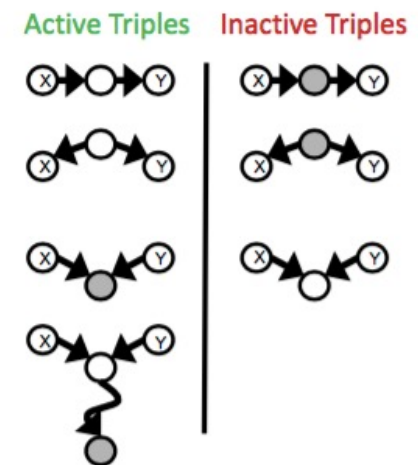
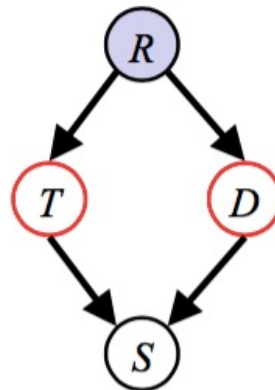
- Questions:

$$T \perp\!\!\!\perp D$$

No

$$T \perp\!\!\!\perp D | R$$

Yes, Independent



Example

- Variables:

- R: Raining
- T: Traffic
- D: Roof drips
- S: I'm sad

- Questions:

$T \perp\!\!\!\perp D$ *No*

$T \perp\!\!\!\perp D | R$ *Yes, Independent*

$T \perp\!\!\!\perp D | R, S$ *No*

